

Administration of a herbal immunoregulation mixture enhances some immune parameters in carp (*Cyprinus carpio*)

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Abstract The herbal immunoregulation mixture (HIRM) were extracts of several traditional Chinese medicines (TCMs): *Astragalus membranaceus* (from the root and stem), *Polygonum multiflorum* Thunb. (from the root), *Isatis tinctoria* L. (from the root), *Glycyrrhiza glabra* (from the stem). Immune parameters, which included macrophage phagocytic activity, macrophage reactive oxygen species (ROS), activity of serum lysozyme, nitric oxide synthetase (NOS) and superoxide dismutase (SOD), levels of total serum protein, globulin, albumin, triglyceride and cholesterol, were determined in carp that had been fed diets containing HIRM at 0.5% or 1% for 30 days. The results showed that, compared with those in the control group, the diets with 0.5% and 1% HIRM resulted in significant increase in macrophage phagocytic activity, macrophage ROS and the levels of total protein, globulin, albumin and NOS activity in serum ($P < 0.05$), and no significant difference was

found in SOD, lysozyme activities and triglyceride level ($P > 0.05$). On the other hand, 0.5% HIRM led to evident enhancement of NOS activity and cholesterol level compared to 1% HIRM. These results indicated that HIRM might elevate the function of immunity in carp (*Cyprinus carpio*).

Keywords Biochemical indices of serum · *Cyprinus carpio* · HIRM · Lysozyme · Macrophage phagocytic activity · NOS · ROS · SOD

Introduction

The impact of antibiotic-resistant bacteria on human health has become a major international concern in recent years. Attention has focused on food-producing animals as one of several potential sources of antibiotic-resistant bacteria in humans. These resistant bacteria may spread from animals to humans via the food chain. They may also transfer their antibiotic-resistance genes into human pathogenic bacteria, leading to failure of antibiotic treatment for some life-threatening conditions. At the same time, other side effects, such as the residual impact on the environment, are also risk factors when commercial antibiotics are used for disease in fish. Novel methods of minimizing risk include control of antibiotics use

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and substitution of antibiotics (Salisbury et al. 2002; Sakai 1999; Gilliver et al. 1999).

In the development of alternative practices for disease management in aquaculture, attention should be diverted to find immunostimulant substances from plant sources that are safe to animals and humans (Joint Expert Technical Advisory Committee on Antibiotic Resistance; JETACAR 1999; Barton 1998). Immunostimulants increase resistance to infectious diseases by enhancing both specific and non-specific defense mechanisms. The immunostimulants that have been applied to aquaculture include glucan (Chen and Ainsworth 1992), lactoferrin (Sakai et al. 1993), chitosan (Siwicki et al. 1994), aloe (Kim et al. 1999) and extracellular products (ECPs) of *Mycobacterium* (Chen et al. 1998). These immunostimulants could improve the immune status of the fish by raising the number of phagocytes and enhancing lysozyme and complement activities and the immunoglobulin level. Some immunostimulants may protect rainbow trout (*Oncorhynchus mykiss*) against diseases such as furunculosis and decrease the mortality of rainbow trout (Wahli et al. 1998). Thus, the use of immunostimulants in fish farming for prevention of fish diseases is a promising method (Raa et al. 1992).

The herbal immunoregulation mixture (HIRM) is a concoction of several traditional Chinese medicines (TCMs): *Astragalus membranaceus*, *Polygonum multiflorum* Thunb, *Isatis tinctoria* L., *Glycyrrhiza* L. et al.

Astragalus polysaccharides (APS), the primary component of the roots of *Astragalus membranaceus*, are a class of macromolecules that can profoundly affect the immune system and therefore have potential as immunomodulators. APS activate mouse B cells and macrophages, in terms of proliferation or cytokine production (Jian and Wu 2003; Tzianabos 2000). *Polygonum multiflorum* Thunb. (Polygonaceae) stems have a history of use in the treatment of rheumatic arthralgia, aching of the loins and knees, and traumatic injuries (Tu et al. 1992). Numerous reports have demonstrated that *Polygonum multiflorum* extract (PME) exhibits a variety of pharmacological effects, such as antioxidative action (Chiu et al. 2002) and free-radical scavenging effects (Chen

et al. 1999). The root of *Isatis tinctoria* L., popularly known as Ban-Lan-Gen, is used in traditional Chinese medicine for seasonal febrile diseases, pestilence, mumps, eruptive diseases, inflammatory diseases with redness of skin, sore throat, etc. A study showed that Ban-Lan-Gen had antibacterial and anti-inflammatory activities (Zhai and Guo 1987). Radix *Glycyrrhiza glabra*, licorice, is an extensively used herbal drug in Eastern medicine. A number of bioactive compounds in licorice have been described, e.g., glycyrrhizin (GL) and its aglycon glycyrrhetic acid (GA), liquiritin (LQ), liquiritin apioside (LA), isoliquiritin (IL) and glabridin (GLAB) (Cinatl et al. 2003).

Traditional Chinese medicines have been used widely to treat human and animal diseases in China (Huang et al. 2004). They have fewer side effects than antibiotics during treatment of diseases. They can profoundly affect the immune system and, therefore, have potential as immunomodulators. The immunostimulant effects of these TCMs have recently been investigated extensively. HIRM is a good source of a large number of bioactive substances and can be used as an immunostimulant. However, application of multifunctional phytochemicals in aquaculture is new, and few studies have been carried out in fish (Jian and Wu 2003). The present study was undertaken to evaluate the effects of HIRM on several immune and biochemical parameters in carp (*Cyprinus carpio*).

Materials and methods

The herbs, including *Astragalus membranaceus*, *Polygonum multiflorum* Thunb., *Aristolochia* L. (Aristolochiaceae), *Isatis tinctoria* L., *Glycyrrhiza glabra*, etc. were washed several times in water and powdered with a pestle in 10% ethyl alcohol (Harikrishnan et al. 2003). The extracts of these herbs (HIRM) were incorporated into a control diet at a rate of 0.5% or 1% (w/w) as experimental diets. The control diet contained 32% crude protein, 5.3% crude lipid, 10% ash, 8% crude fiber, 1.2% calcium and 0.4% phosphorus. No other immunostimulant was added in the control diet.

The carp were obtained from a freshwater fish farm at the Freshwater Fisheries Research Institute of Jiangsu Province. Weighing 80 ± 1.5 g, they were divided equally into three groups with 100 fish in each group. They were acclimated in indoor cement ponds ($9 \text{ m} \times 9 \text{ m} \times 2 \text{ m}$), which received continuous aeration, and 50% of the water was exchanged weekly to maintain water quality. During the grow-out periods, water temperature ranged from 23°C to 28°C , pH from 6.91 to 7.44, salinity from 10‰ to 20‰, and dissolved oxygen concentration from 6.63 mg/l to 7.78 mg/l. First, all the fish were fed the control diet for 20 days. After acclimatization, the experimental groups received experimental diets (0.5% or 1% HIRM in the diets), and the control group was fed with control diet for 30 days at the same time. They were fed at 8:00 h, 13:00 h and 21:00 h daily during husbandry at a rate of 3% of body weight following assessment of biomass by bulk weighing every 7 days.

Blood was sampled from the caudal vein of an experimental fish. The blood was allowed to clot at room temperature. Serum was obtained by centrifugation at 400 g and stored at -20°C for assaying activity of lysozyme, nitric oxide synthetase (NOS) and superoxide dismutase (SOD), and levels of total protein, globulin, albumin, triglyceride and cholesterol (Rao and Chakrabarti 2005).

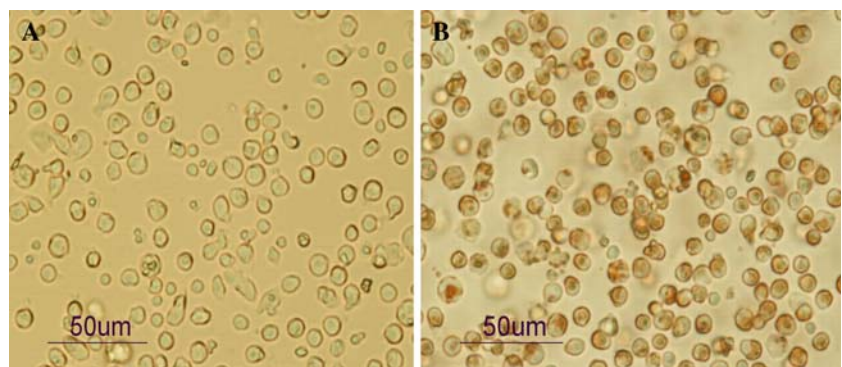
After blood collection, the head kidney was removed from the fish for macrophage phagocytic assay and reactive oxygen species (ROS) assay. Macrophages were isolated from the kidney according to the method of Secombes (1990). The head kidney of the fish was transferred to 10 ml of 1640 medium supplemented with 5%

fetal calf serum (FCS). We prepared a cell suspension by grinding the kidney and filtering it through a stainless steel mesh (diameter 0.3 mm) in a Petri dish on ice. The suspension was layered onto a 34%/51% (v/v) Percoll gradient and centrifuged for 25 min at 400 g at 4°C . The macrophage-enriched fraction was collected, and the cells were counted in a hemocytometer. The cells were washed in 1640 medium twice by centrifugation. The supernatant was discarded, and the pellet was resuspended in 1640 medium containing 5% FCS at a concentration of 5×10^5 cells ml^{-1} .

The macrophage phagocytic activity was studied by the method of Yang et al. (2004). Macrophages from the head kidney were incubated in 96-well plates (100 μl each well) at 27°C for 1 h. After incubation, 10 μl neutral red (0.072%) was added to the wells for phagocytizing for 30 min (Fig. 1); the cells were washed three times with Hank's balanced salt solution (HBSS), and 200 μl ethanol and acetic acid (50:1) solution was added to each well. The optical density (OD) was then measured at 540 nm with a fluorescence microreader (Bio-TEK, Highland Park, Vermont, USA).

The ROS of carp macrophages was studied by the method of Bayne and Levy (1991). Inside the cells, 2',7'-dichlorodihydrofluorescein diacetate (DCFH-DA; Sigma-Aldrich, USA) was cleaved by esterases, forming DCFH, which was in non-fluorescent form and was oxidized to fluorescent compound DCF by ROS. Of the macrophage suspension [5×10^5 cells/ml in Roswell Park Memorial Institute (RPMI) medium 1640], 100 μl was placed in a 96-well microtiter polystyrene plate at 27°C for 1 h. Non-adhered cells were removed by three washes with HBSS. Fifty

Fig. 1 **A** macrophages under normal conditions. **B** macrophages after being phagocytized in neutral red and washed three times with HBSS. The photographs were taken under fluorescence microscopy (Nikon ECLIPSE TE2000-S, Tokyo, Japan)



microliters of phorbol myristate scetate (PMA; Invitrogen, USA) solution (50 ng/ml) and 50 µl DCFH-DA solution (1×10^{-5} mol/l) were placed in the wells of a 96-well microtiter plate containing a macrophage monolayer. The plate was shaken, and cellular fluorescence intensity was detected with a fluorescence microreader (BIO-RAD, Model 550 microplate reader) at excitation length 485 nm and emission length 528 nm.

Carp serum lysozyme activity was determined with a lysozyme kit (Jiancheng Bioengineering Institute, Nanjing, China). According to the manufacturer's instructions, 200 µl of fish serum sample was added to 1,800 µl suspension of *Micrococcus lysodeikticus* (0.2 mg/ml) in sodium phosphate buffer. The reaction was carried out at 37°C for 15 min, and the suspension was measured at 530 nm in ten replicate wells with 200 µl in each well (96-well microtiter plate) by the fluorescence microreader. Lysozyme activity was defined as micrograms per milliliter.

We measured serum SOD activity by its ability to inhibit superoxide radical-dependent reactions, using the Jiancheng SOD Kit (Jiancheng Bioengineering Institute). SOD activity was determined by hydroxylamine assay, developed from a xanthine oxidase assay. Briefly, the xanthine and 2-(4-iodophenyl)-3-(4-nitrophenol)-5-phenyltetrazolium chloride (INT) was dissolved in CAPS and EDTA. In the presence of xanthine oxidase, superoxide was produced from xanthine. The superoxide radical then reacted with INT to produce a red formazan dye. The OD value was measured at 550 nm after reaction for 40 min. A reference standard SOD was supplied with the Jiancheng SOD Kit. SOD activity was expressed as SOD units per milliliter of serum.

NOS catalyzes L-arginine and O_2 to produce nitric oxide (NO). The NO reacts with nucleophilic substances to form a colored compound that assays NOS activity. We measured this activity using the Jiancheng SOD Kit (Jiancheng Bioengineering Institute). The OD was measured at 530 nm.

The biochemical indices were determined with a Hitachi biochemical analysis instrument (7600). These indices included total protein, albumin, globulin, triglyceride and cholesterol. They were determined spectrophotometrically in the UV area.

Each experiment was repeated three times. Data are presented as means $\pm$ SDs ($n = 10$) of fish per group. All the immune parameters were analyzed by Student's *t*-test to compare the differences in values between control and experimental fish. Differences were regarded as significant when $P < 0.05$.

Results

Assay of macrophage phagocytic activity and ROS

Macrophage phagocytic activity and ROS level are shown in Fig. 2. After the fish had fed for 30 days, the macrophage phagocytic activities at an OD of 540nm were 2.03 ± 0.1 , 2.75 ± 0.29 and 2.57 ± 0.05 in the control group, 0.5% and 1% HIRM, respectively. The ROS levels at 485/528nm were 136.36 ± 4.66 , 171 ± 7.47 and 153.86 ± 6.9 in the control group, 0.5% and 1% HIRM, respectively.

Assay of lysozyme, SOD and NOS activity

The activities of serum lysozyme, SOD and NOS after the fish had been fed for 30 days are shown in Fig. 3. The lysozyme activities were 145.62 ± 5.56 µg/ml, 158.98 ± 8.16 µg/ml and 129.82 ± 8.37 µg/ml in the control group, 0.5% and 1% HIRM, respectively. The SOD activities were 148.3 ± 8.12 units/ml, 137.4 ± 5.44 units/ml and 148.22 ± 1.83 units/ml in the control group, 0.5% and 1% HIRM, respectively. The NOS activities were 29.07 ± 3.45 units/ml, 32.68 ± 3.67 units/ml and 29.9 ± 4.23 units/ml in the control group, 0.5% and 1% HIRM, respectively.

Assay of serum total protein, albumin and globulin levels

Total protein, albumin and globulin levels are shown in Fig. 4. The total protein levels were 23.84 ± 3.72 g/l, 28.56 ± 3.01 g/l and 30.36 ± 3.07 g/l in the control group, 0.5% and 1% HIRM, respectively. The albumin levels were 10.35 ± 1.61 g/l, 12.21 ± 1.26 g/l and 12.44 ± 1.32 g/l in the control group, 0.5% and 1%

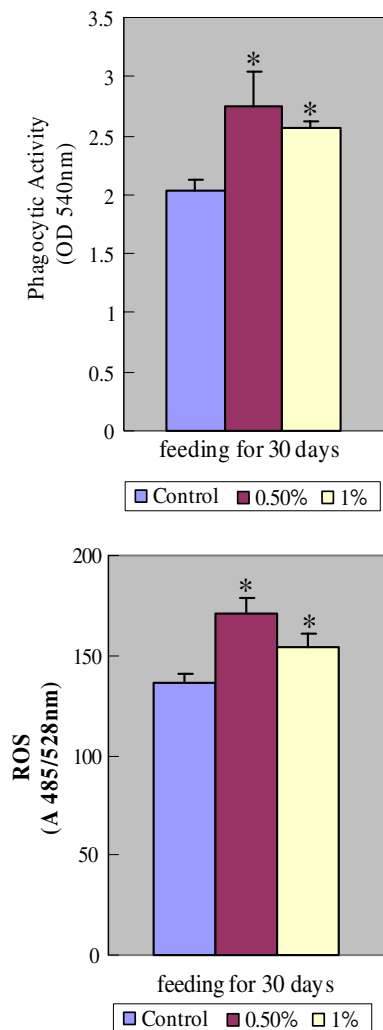


Fig. 2 Phagocytic activity and ROS level in control and experimental groups. The values represent the means $\pm$ SDs ($n = 10$). An asterisk denotes a significant effect of dietary treatment as determined by Student's t -test (* $P < 0.05$)

HIRM, respectively. The globulin levels were 13.49 ± 1.16 g/l, 17.35 ± 1.95 g/l and 17.93 ± 1.94 g/l in the control group, 0.5% and 1% HIRM, respectively.

Assay of serum triglyceride and cholesterol levels

The serum triglyceride and cholesterol levels are shown in Fig. 5. The triglyceride levels were 2.73 ± 0.74 mmol/l, 3.09 ± 0.64 mmol/l and 3.17 ± 0.47 mmol/l in the control group, 0.5%

and 1% HIRM, respectively. The cholesterol levels were 2.99 ± 0.75 mmol/l, 3.78 ± 0.5 mmol/l and 3.1 ± 0.65 mmol/l in the control group, 0.5% and 1% HIRM, respectively.

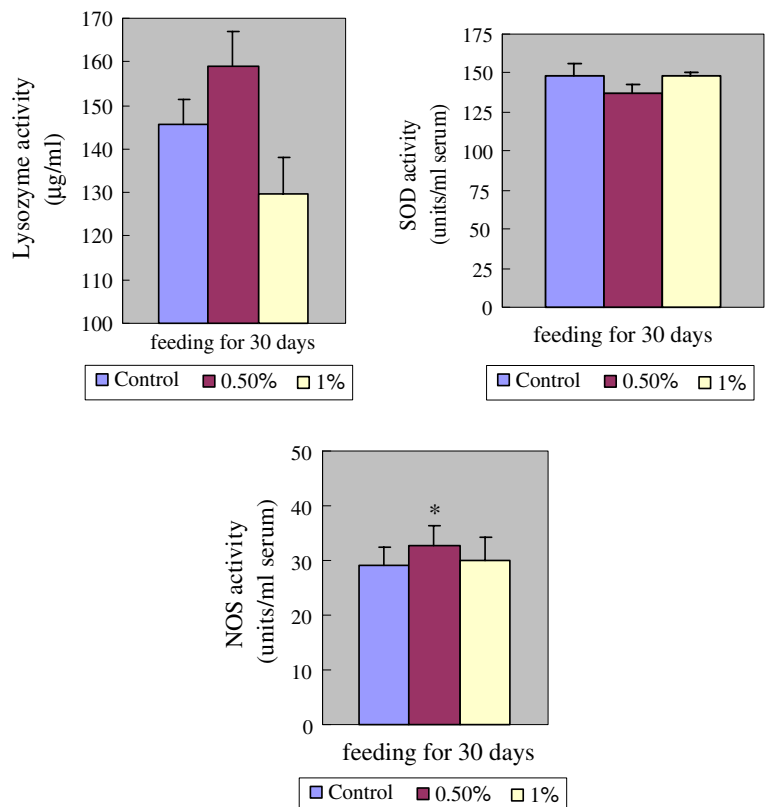
Discussion

Studies showed that oral administration of quil-laja saponin enhanced the immunity of yellowtail (Ninomiya et al. 1995), and feeding with *Catharantus roseus* plant extract enhanced the immune response of *Labeo rohita* (Nguyen et al. 2002). Administration through injection enabled the immunostimulant to be quickly absorbed and functional, while in the oral administration, the immunostimulant is slowly absorbed by the fish (Yoshida et al. 1995). However, the drawback of injection is that it is labor intensive and also stressful to the fish. Thus, oral administration is a useful alternative method for mass administration to fish of all sizes (Selvaraj et al. 2005).

The innate immune system is the only defense weapon of invertebrates and a fundamental defense mechanism of fish (Magnadottir 2006). The ability of macrophages to kill pathogenic microbes is probably one of the most important mechanisms of protection. Since $O_2^{\cdot-}$ (superoxide anion) is the first product released during the respiratory burst, $O_2^{\cdot-}$ (superoxide anion) concentration has been accepted as an accurate parameter to quantify the intensity of a respiratory burst (Secombes 1990). The phagocytic cells might have retained the capacity to produce superoxide anion radicals. This is important for mediating immunity. Reactive oxygen species appear to be important in some signaling pathways involved in phagocyte activation (Schreck et al. 1992). Thus, the macrophage generated reactive oxygen species could exert an important control on the steady state concentration of these reactive oxygen species and, consequently, the control of a specific immune response. This paper showed that the macrophage phagocytic activity and ROS level in the 0.5% and 1% HIRM groups were significantly increased compared with the control group ($P < 0.05$).

As a first line of defense, various peptides, such as lysozyme, complement factors, globulin and

Fig. 3 Activity of lysozyme, SOD and NOS in control and experimental groups. The values represent the means $\pm$ SDs ($n = 10$). The *asterisk* denotes a significant effect of dietary treatment as determined by Student's *t*-test ($*P < 0.05$)



other lytic factors, are present in serum, where they prevent adherence and colonization of microorganisms (Alexandar and Ingram 1992), leading to prevention of infection and disease. Also, lysozyme contributes to the innate immunity of animals by their bactericidal and anti-inflammatory properties (Jollés and Jollés 1984). Results demonstrated that there were marked changes in serum total protein, albumin and globulin with HIRM administration for 30 days. The results showed that, compared with those in the control group, total protein, albumin and globulin levels in the 0.5% and 1% HIRM groups were all significantly increased in serum ($P < 0.05$), but no significant difference was found in lysozyme ($P > 0.05$).

Activated macrophages express NOS and then increases NO synthesis. NO plays a dual role in the inflammatory process and has been shown to be involved in numerous regulatory functions, ranging from altering the cardiovascular system to modulating neuronal functions. SOD reacts with $O_2^{\bullet -}$ at rates limited only by diffusion and

enhanced by electrostatic guidance (Fridovich 1995; Getzoff 1992), providing a highly effective means of removing $O_2^{\bullet -}$. Our paper indicates that the NOS activities had significantly increased in the 0.5% HIRM group compared to the control ($P < 0.05$), but 1% HIRM was almost similar to the control ($P > 0.05$). In addition, in SOD activity, no significant differences were detected between the 0.5% and 1% HIRM groups and the control group ($P > 0.05$).

In fact, nutritional and immunological conditions are closely related. In the present study, these data showed that the cholesterol level had significantly increased in the 0.5% HIRM group compared to the control group ($P < 0.05$), but the cholesterol level of the 1% HIRM group was almost similar to that of the control. The triglyceride levels were slightly higher in the 0.5% and 1% HIRM groups than in the control, but there was no significant difference ($P > 0.05$).

In conclusion, our study has shown that HIRM treatment might have an interesting, potential efficacy on immunity in carp. HIRM is able to

Fig. 4 Total protein, albumin and globulin levels in control and experimental groups. The values represent the means $\pm$ SDs ($n = 10$). An *asterisk* denotes a significant effect of dietary treatment as determined by Student's *t*-test ($*P < 0.05$)

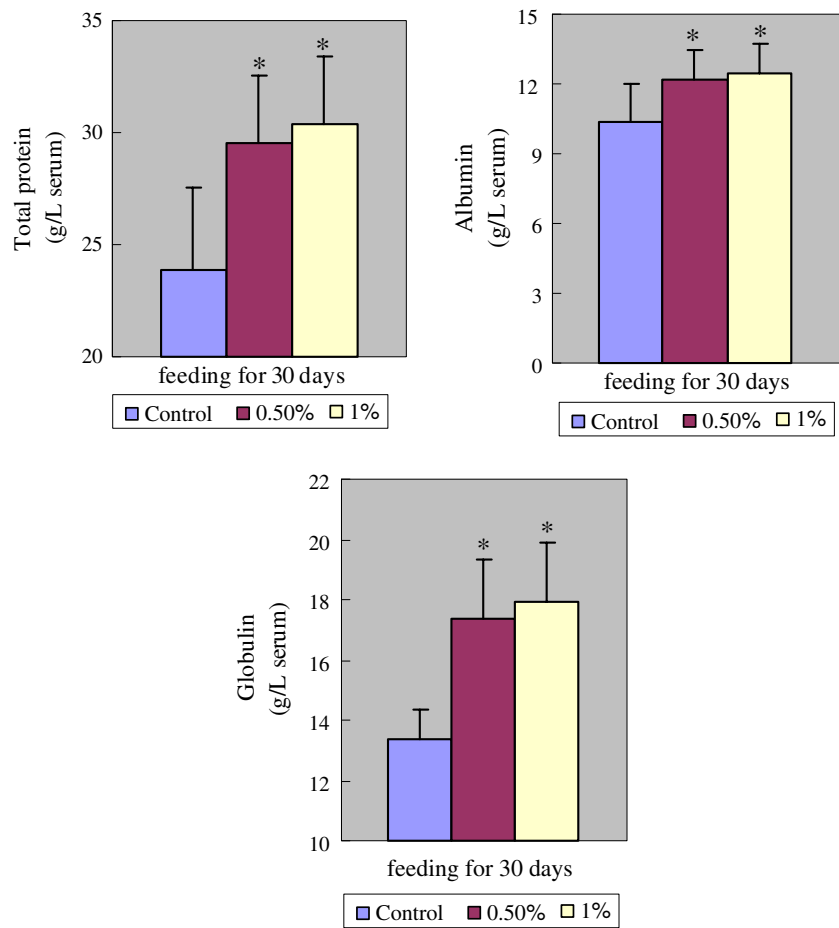
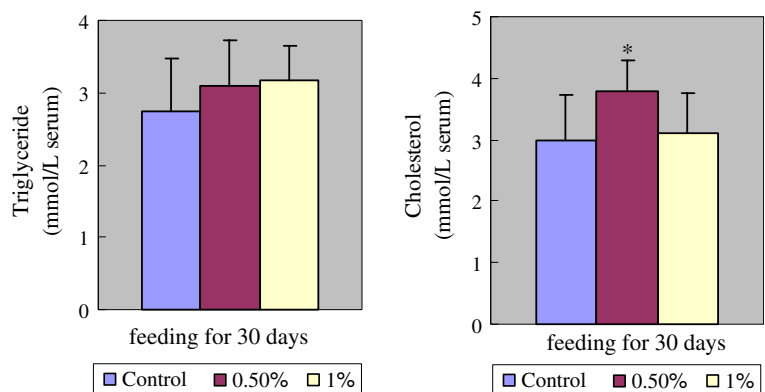


Fig. 5 Triglyceride and cholesterol levels in control and experimental groups. The values represent the means $\pm$ SDs ($n = 10$). The *asterisk* denotes a significant effect of dietary treatment as determined by Student's *t*-test ($*P < 0.05$)



modulate some immune parameters in carp, *Cyprinus carpio*. The effect of HIRM by oral delivery should be further investigated in relation to a molecular mechanism to evaluate its efficacy in increasing disease resistance.

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